V-L-51c
April 26, 1968
SUPERSEDING
Fed. Spec. V-L-51b
July 2, 1964

FEDERAL SPECIFICATION

LACES, COTTON

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. The specification covers the requirements for cotton laces.

1.2 Classification.

1.2.1 Types and classes. The laces shall be of one type in the following classes, as specified (see 6.2).

Type I - Mercerized.

Class A - Deleted (see 6.5).
Class B - 7/32 inch width (64 carrier).
Class C - 7/32 inch width (44 carrier).

Type II - Deleted (see 6.5).

Type III - Deleted (see 6.5).

Type IV - Deleted (see 6.5).

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

UU-P-268 - Paper, Kraft, Untreated Wrapping.
CCC-T-191 - Textile Test Methods.
PPP-B-636 - Box, Fiberboard.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

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FSC 8335

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Samples

3.1.1 Guide sample. Guide samples, when furnished, are solely for guidance and information to the supplier (see 6.3). Variation from this specification may appear in the sample, in which case the specification shall govern.

3.1.2 Standard sample. When a standard sample is established, the finished laces shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.1.3 Preproduction sample. When specified (see 6.2), before production is commenced, twenty finished laces shall be submitted or made ready for the contracting officer or his authorized representative for inspection as specified in 4.2. The approval of the preproduction sample authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Material.

3.2.1 Cotton yarn. The cotton yarn used in the manufacture of the laces shall be mercerized and shall be in the size and ply specified in table I.

TABLE I. Physical requirements for type I laces

Class	Width (± 1/32 tolerance)	Yarn size, ply (minimum)	Ends per carrier (minimum)	Number of carriers (minimum)	Picks per inch (minimum)	Breaking strength (minimum)
	Inch					Pounds
B	7/32	40/2	1	64	38	70
C	7/32	50/2	2	44	26	70

3.2.2 Tips. The tips shall be made from nylon sheet or film having a thickness of 0.013 to 0.016 inch. The color of the tips shall be black for black laces and clear for all other color laces.

3.3 Color. The color of the finished laces shall be as specified (see 6.2). Unless specifically authorized, the use of sulfur dyes and dyes containing elementary sulfur or compounds capable of oxidation to sulfuric acid is prohibited. The dyed laces shall show no more labile sulfur than shown on the standard sample when tested as specified in 4.3.2.2.

3.3.1 Matching. The color shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500°Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2700°Kelvin.

3.3.2 Colorfastness. The dyed laces shall show colorfastness to water equal to or better than the standard sample when tested as specified in 4.3.2.2. When no standard sample has been established the dyed laces shall show good colorfastness to water when tested as specified in 4.3.2.2.

3.4 Braid. The braid for the laces shall be tubular braided without a core using two over and two under braiding. The braid shall be pressed flat and shall conform to the requirements in table I when tested as specified in 4.3.2.2.

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3.5 Tipping. Both ends of the lace shall be provided with a firmly attached and secured tip made from the material specified in 3.2.2. Tips shall be 3/4 ± 1/16 inch long.

3.5.1 Gripping strength. The tips shall have a minimum gripping strength of 20 pounds when tested as specified in 4.3.2.2.

3.6 Length. Laces shall be furnished in lengths, including tips, as specified (see 6.2). The laces shall be cut to the specified length ± 1/2 inch, when tested as specified in 4.3.2.2.

3.7 Workmanship. The finished laces shall conform to the quality and grade of product established by this specification. Occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Preproduction sample inspection. When required, the preproduction sample submitted in accordance with 3.1.3 shall be visually inspected for appearance, color and finish. The sample shall be tested for physical and chemical characteristics in accordance with 4.3.2.2 to the extent applicable.

4.3 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents. In addition, testing shall be performed on the component listed in table II for the characteristics shown. The inspection level for the component shall be level S-1 and the acceptable quality level (AQL) for the component shall be 4.0 test failures per hundred units. All requirements are applicable to the sample unit.

TABLE II. Component tests

Component and lot expressed in terms of	Character-istic	Requirement paragraph	Test method	No. det. per sample unit	Results reported as	Sample unit
Nylon sheet (10 sheets)	Thickness	3.2.2	Gauge	3	Ave. of 3 det. to the nearest 0.001 inch	6 by 1-1/2 inch piece or equivalent
	Material identifi-cation	3.2.2	1/	1	Pass or fail	

1/ The supplier shall submit a certificate of compliance for this characteristic.

4.3.2 Inspection of the end item.

4.3.2.1 Examination of the end item. The laces shall be examined for defects in workmanship, material, overall length, width, and overall length of tips. Defects found during these examinations shall be classified in accordance with 4.3.2.1.1, 4.3.2.1.2 and 4.3.2.1.3. The sample unit for these examinations shall be one finished lace and the lot size shall be expressed in units of one lace each.

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4.3.2.1.1 Examination of laces for defects in material and workmanship. The inspection level for this examination shall be level I and the AQL shall be 1.0 defects per hundred units.

Examine	Defect
Laces	Any broken yarn. Frayed. Rub or abrasion mark. Hole, cut or tear. Unevenly braided resulting in open place or break in continuity of braid. Pulled yarn or uneven tension resulting in uneven width. Other than tubular braided without a core. Not flat for a distance of 2 inches or more. Slub, knot, or abruptly thickened place which may affect utility. Color other than specified. Off shade. Spot, stain, or other discoloration affecting appearance. Objectionable odor.
Tips	Any tip missing, damaged, malformed or insecurely attached. Tipping of braid not neatly accomplished. Color not black for black laces. Color not clear for laces other than black. End of tip projects beyond the braid more than 1/16 inch. End of braid projects beyond the tip end more than 1/16 inch.

4.3.2.1.2 Examination for width and length. Any width measured in accordance with 4.4.1 that exceeds the tolerance in table I, or any length measured in accordance with 4.4.2 that exceeds the tolerance specified in 3.6 shall be classified as a defect. The inspection level shall be S-3 and the AQL shall be 1.0 defect per hundred units.

4.3.2.1.3 Examination for overall length of tips. Any overall length of tip exceeding specified tolerances indicated in 3.5 shall be classified as a defect. The inspection level shall be S-3 and the AQL shall be 1.0 defect per hundred units.

4.3.2.2. Testing of the end item. Testing of the completely fabricated laces shall be performed in accordance with CCC-T-191 and table III for characteristics shown therein. The inspection level shall be S-1 and the AQL shall be 2.5 test failures per hundred units. The sample unit shall be twenty completely fabricated laces. Requirements are applicable to the sample unit. When the data in the "Number of determinations per sample unit" and "Results reported as" columns are not specified in table III, they shall be expressed in units of one lace each.

TABLE III. End item tests

Characteristic	Requirement paragraph	Test method	No. of determinations per sample unit	Results reported as
Cotton identification	3.2.1	1200 1/	-	-----
Labile sulfur	3.3	2020 1/	1	Pass or fail
Colorfastness to water	3.3.2	4630	-	-----
Braiding type	3.4	Visual	1	Pass or fail
Yarn size	3.2.1	4020	-	-----
Yarn ply	3.2.1	Visual	1	Number of plies
Ends per carrier	Table I	Visual	1	Number of ends
Carriers	Table I	Visual	1	Number of carriers
Picks per inch	Table I	visual 2/	3	Avg. of 3 det. to nearest whole number
Breaking strength	Table I	4.4.3	-	-----
Gripping strength (tips)	3.5.1	4.4.4	-	-----

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1/ The supplier shall submit a certificate of compliance for this characteristic.

2/ A pick glass or suitable magnifying glass and scale may be used.

4.3.3 Examination of preparation for delivery requirements. An examination shall be made to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery, with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. the inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

Examination	Defects
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, improper taping, loose strapping, inadequate stapling. Buldged or distorted container.
Content	Number of laces per bundle is more or less than required. 1/
Weight	Weight of the contents of container exceeds the maximum requirement.

1/ One bundle from each shipping container in the sample shall be examined.

4.4 Tests.

4.4.1 Width. The width shall be determined at 3 points at least 9 inches apart, using a steel rule graduated in thrity-seconds of an inch or a caliper type gauge, and the results averaged. The measurements shall be taken with the test lace suspended vertically under a tension of 3 ounces.

4.4.2 Length. The finished lace shall not be stretched or tensioned prior to the length test. The finished lace shall be suspended vertically under a tension of 3 ounces and then measured from end to end including the tips.

4.4.3 Breaking strength. The breaking strength of the laces shall be determined as specified in method 4102 of CCC-T-191, except that the clamps shall be not less than 2 inches in diameter and the distance between the center of the clamps at the start of the test shall be 6 ± 1/16 inches. When the lace length is less than 24 inches, the flat type jaw shall be used in lieu of a drum type jaw.

4.4.4 Gripping strength of tips.

4.4.4.1 Test specimen. The specimen shall be 1/2 the length of a lace. (Two determinations may be made from one complete lace.)

4.4.4.2 Apparatus.

4.4.4.2.1 Testing machine. A testing machine as described in method 5100 except that a slotted metal plate, as shown in figure 1, shall be used in connection with the top jaw assembly and the distance between the upper jaw (slotted metal plate) and the lower jaws or clamps shall be 4 ± 1/16 inches. The lower jaw shall be as described in method 5100 or 4102 of CCC-T-191.

4.4.4.2.2 Metal plate (figure 1). The rigid metal plate shall be approximately 5 inches long by 1-1/2 inches wide and shall contain a slot 4 inches long centered widthwise and tapering lengthwise from a dimension of 1/4 inch in width at the outside edge to a dimension of 1/32 inch at the inner end. The slot shall be smoothly finished. It is not required that the plate be of a single piece of metal. The slot may be formed by a clamped or bolted assembly of two metal bars or plates properly spaced apart. The plate shall be supported in a horizontal position in or on top jaw of the apparatus.

4.4.4.3 Procedure. The lace shall be passed upward, tip first, through the jaw assembly and through the wide end of the slot until the tip clears the plate, and then moved horizontally toward

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the narrow end of the slot until the end of the tip rests on top of the metal plate. The free end of the specimen (lace) shall be fastened in the lower jaw or clamp. The machine shall be started, with the pawl in contact with the ratchet, and the maximum resistance in displacing the tip shall be recorded. The force shall be applied to the specimen at such a rate that the pulling clamp will travel at a uniform speed of 12 ± 0.5 inches per minute. After the tip has been forced from the lace, the resultant force shall be read from the dial, scale or chart and the value recorded to the nearest pound. If the lace tip slips through the slot without its being removed from the lace or if for any reason attributed to faulty technique, an individual measurement falls markedly below the average test for the sample unit, such result shall be discarded and another specimen shall be tested. Buckling of the tip material at the base of the tip without the tip being removed from the lace shall not be considered as a failure of the tip.

4.4.4.4 Report. Ten specimens shall be tested from each sample unit. The gripping strength of the sample unit shall be the average of the results and shall be reported to the nearest 1.0 pound.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packing shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Five gross of laces of one description only shall be put up in a compact bundle. Prior to bundling, the laces shall be grouped full length in one gross lots and securely tied together. Each bundle of laces shall be completely wrapped in 60-pound basis weight kraft paper conforming to grade B of UU-P-268. The wrap shall be securely sealed with 2-inch minimum width gummed paper tape conforming to type III, grade B of PPP-T-45.

5.1.2 Level C. Laces shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A,B, or C as specified (see 6.2).

5.2.1 Level A. Laces of one description only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a taped liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-F-320. Each shipping container shall be closed, waterproofed and reinforced with strapping or tape banding in accordance with the appendix of the container specification. The weight of the contents of each shipping container shall not exceed 65 pounds.

5.2.2 Level B. Laces of one description only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a taped liner conforming to class domestic, variety DW, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of the container specification. The weight of the contents of each shipping container shall not exceed 65 pounds.

5.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c or V3s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the container specification. The shipping container material may also be grade V4s of PPP-B-320.

5.2.3 Level C. Laces, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery to destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special markings required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The cotton laces covered by this specification are intended for use in footwear and other items.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Class and length required (see 1.2.1 and 3.6).
(c) When preproduction sample is required (see 3.1.3).
(d) Color required (see 3.3).
(e) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(f) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 Samples. For access to samples, address the procuring office issuing the invitation for bids.

6.4 Civil agency procurement. When level B packaging requirements are specified for civil agency procurements, the requirements in 5.1.1 shall apply.

6.5 Type I, class A; type II, classes A and B; type III, classes A and B; type IV, classes A, B, C, and D have been deleted. Type I, classes B and C have been retained.

6.6 Information is furnished that the artifical light requirements in 3.3.1 have been met by the Macbeth Super Skylight BX 1014 which may be purchased from the Macbeth Daylight Corporation Box 950T, Newburgh, N. Y.

MILITARY CUSTODIANS:

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Navy - SA
Air Force - 82

Review activities:

Navy - SA
Air Force - 11, 82, 85

User activity:

Army - MD

Preparing activity:

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CIVIL AGENCIES INTEREST:

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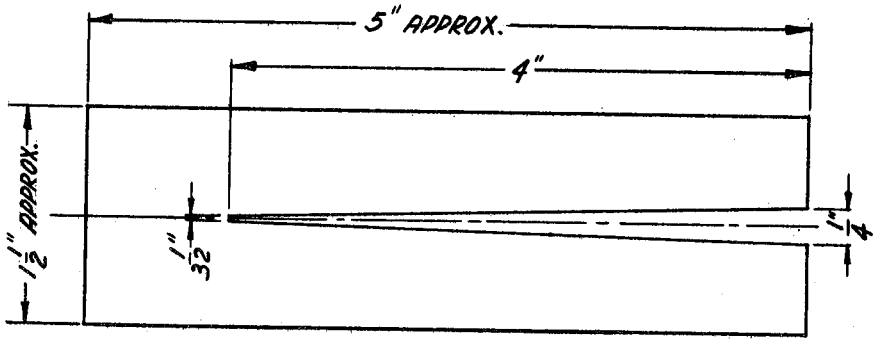


FIGURE 1. Slotted metal plate.

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

Laces, Cotton - V-L-51c

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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